Cheerio Product Intern Assignment — Assignment A

Scaling Customer Support for Non Technical Users | Candidate Submission

Part 1: Solution

1. User Persona & The Top 5 Setup Blockers

- **Target User:** Aman Mishra, founder of **GATESprint Academy**, an online coaching institute in India preparing students for GATE exams (CSE, DA, ME, EE, EC). Aman captures student enquiries via Instagram/Facebook ads and converts them over WhatsApp. He is non technical and gets anxious about routine setup failures right before admissions deadlines.
- **Top 5 Setup Blockers (in chronological order of friction):**

1. **Connecting WhatsApp Business Number:** WABA registration, two-factor authentication, and Meta display name verification. Without this, no campaigns can run.
2. **Meta Template Rejections:** Automated rejections by Meta's compliance filters due to perceived promotional hype (e.g., "guaranteed rank boost").
3. **Building the Student Qualification Flow:** Structuring interactive questionnaires (GATE stream, exam year, prep status) without complex workflow logic.
4. **Google Sheets Sync & Counsellor Routing:** Mapping student responses to spreadsheets and alerting counsellor Priya Sharma without broken webhooks.
5. **Campaign Delivery Troubleshooting:** Diagnosing why broadcast messages show as "Pending" or "Failed" before time-sensitive weekend webinars.

2. Solution Prioritization

To scale support, I evaluated 4 potential approaches against non technical user realities:

Solution Option	Easy for Non Technical?	Creates the Setup?	Fixes Root Cause?	Verdict
1. Help Docs & Videos	Low (requires reading)	No (manual work)	Partial	**Rejected:** Users under deadline stress skip docs and open tickets.
2. Static Checklist Wizard	Medium (step-by-step)	Partial (rigid steps)	No (generic)	**Rejected:** Too generic; cannot write copy or adapt to GATE streams.
3. Rule-Based Troubleshooter	Medium (decision tree)	No (reactive only)	Yes (for known bugs)	**Rejected:** Purely reactive; only triggers after something breaks.
4. AI Setup Copilot	**High (natural language)**	**Yes (drafts copy/forms)**	**Yes (pre-flight checks)**	**Selected: Solves both creation and real-time error prevention.**

3. Chosen Solution: Cheerio Setup Copilot

An in-product assistant that translates everyday language (*"I want to run a free GATE strategy session"*) into live, verified WhatsApp workflows.

- **What It Does:** Interprets campaign goals, auto-populates configuration fields, drafts Meta-compliant templates, structures qualification questions, connects Google Sheets, and pre-screens for errors.
- **What It Will Not Do in V1:** It will not auto-publish without human review, handle high-risk Meta policy appeals, manage billing disputes, or resolve custom backend webhooks.

4. Human-in-the-Loop Safeguards & Escalation

6. **User Sign-Off Gate:** The Copilot drafts the setup on a live canvas, but the live broadcast button remains disabled until Aman reviews the diff and checks an explicit confirmation box.
7. **Support Handover (for the 40% edge cases):** If an issue cannot be resolved within 3 turns, Cheerio packages the full canvas state, error codes, and transcript for instant human agent takeover without repeating information.

5. Metrics & North Star Target

- **Metric 1: Setup Completion Rate** (% of started setups successfully published).
- **Metric 2: Time-to-First-Broadcast** (reduction in onboarding friction from 48h to <15 mins).

- **Metric 3: CSAT Micro-Score** (1-click thumbs up/down post-session helpfulness).
- **North Star Metric: 60% Self-Serve Resolution Rate within 3 months** (60 out of 100 setup blockers resolved completely without human agent intervention).

Part 2: Design Prototype

Prototype Architecture: Split-Screen "Chat × Form" Workspace

Rather than a disconnected floating chatbot popup, the prototype uses an integrated split-screen workspace:

- **Left Panel (Conversational Copilot):** Aman chats in plain English, receives contextual questions, inspects pre-flight warnings, and uses quick-action chips.
- **Right Panel (Live Dynamic Canvas):** Renders real interactive form inputs (`<input>`, `<select>`, `<textarea>`) that update in real time as the conversation unfolds. Aman can directly edit any field.
- **Review & Sign-Off Bar:** Sticky footer enforcing human verification before publishing.

End-to-End User Journey (Aman's Scenario)

8. **Goal Input:** Aman types: *"I want to run a free GATE strategy session for 2027/28 aspirants."*
9. **Auto-Configuration:** Copilot structures eligible GATE streams and session time (Sunday 7 PM) into the right-hand canvas.
10. **Qualification Flow:** Copilot structures a 4-question WhatsApp flow (Paper, Year, Prep Level, Callback preference).
11. **Edge Case Handled (Meta Compliance Warning):** Copilot drafts message copy but catches a policy risk: *"The phrase 'get a winning plan' has an 82% rejection risk under Meta rules."* Suggests replacing with *"get a study plan and session details."* Aman applies fix with one click.
12. **CRM Routing & Launch:** Copilot links registrations to Google Sheet `GATESprint Leads` and assigns counsellor Priya Sharma. Aman reviews, checks verification, and hits **Publish Funnel Live**.

- **Live Prototype Link:** `https://batman00723.github.io/Product-Intern-Assignment/`
- **GitHub Repository:** `https://github.com/batman00723/Product-Intern-Assignment`
- **File Reference:** `cheerio-copilot-workspace.html` / `index.html` (Standalone clickable prototype).

Part 3: How I Used AI

Tools Used

I used **Antigravity** as an agentic partner to brainstorm technical edge cases, test prompt iterations, and write the initial interactive HTML/CSS/JS prototype code.

Prompt Log: 3 Effective Prompts & 1 Failed Prompt

13. **Persona & Blockers (Worked Well):** *"I am working on a product assignment for Cheerio (https://cheerio.in). The user is Aman Mishra, who runs GATESprint Academy... List the 5 most common setup issues he will run into in order."*

Outcome: Grounded output in Indian coaching realities; generated the 5 chronological blockers.

14. **Compliance & Edge Cases (Worked Well):** *"Write a WhatsApp broadcast template for an online coaching institute... tell me what phrases Meta usually rejects during review for education businesses."*

Outcome: Surfaced Meta's ban on promotional hype phrases, providing the exact edge case used in the prototype.

15. **Prototype Layout (Worked Well):** *"Create a single HTML file with two columns. On left, chat assistant. On right, form with inputs for campaign name, date, student questions, and sheet routing..."*

Outcome: Generated clean split-screen layout with real interactive form fields.

16. **Support Scaling (Failed & Refined):** *"How can Cheerio scale customer support for non technical users?"*

Why it failed: Yielded textbook SaaS advice (build Intercom docs, hire CSMs).

Refinement: Narrowed prompt to in-product natural language guidance and auto-filled forms so users never have to leave the screen to read docs.

Weaknesses Caught & What I Did Without AI

- **Weaknesses Caught:** (1) AI initially proposed auto-publishing campaigns without human review; I killed this and added the mandatory verification sign-off bar. (2) AI wrapped cards in a flex container that squished into thin unreadable strips at 100% zoom; I fixed this by switching to natural block scrolling with visible inputs.
- **Authored Without AI:** GATESprint case context, Indian admissions counselling workflows, solution trade-off evaluation matrix, human escalation packaging, and the entire unit economics and freemium pricing strategy.

Part 4: Numbers — Cost, Billing, Audit & Savings

1. Volume Guesstimate & Resolution Rate

- **Cheerio Active Businesses:** 5,000 businesses (given).
- **Monthly Support Query Volume:** Blended rate of **1.0 query/business/month** across active campaign creators and maintenance accounts = **5,000 queries/month**.
- **Copilot AI Resolution (60% North Star Target): 3,000 queries/month** resolved self-serve.
- **Escalated to Human Support (40% Edge Cases): 2,000 queries/month**.

2. Running Cost & Groq Token Economics

- **Model Selected:** Llama 3 70B Enterprise hosted on **Groq** (chosen for sub-second inference speed to eliminate UI latency).
- **Pricing & Benchmark:** $0.00059 / 1k input tokens, $0.00079 / 1k output tokens (Groq Enterprise). Live rate: **1 USD ≈ ₹95.76**.
- **Token Usage per Session:** 3,000 input tokens (concise system prompt, account state, pruned history) + 250 output tokens (structured suggestions and JSON updates).
- **Cost per Query:** $\frac{3000}{1000} \times \$0.00059 + \frac{250}{1000} \times \$0.00079 = \mathbf{\$0.0019675}$ (~**₹0.19 per query**).
- **Total Monthly AI Running Cost:** $5,000 \times \$0.0019675 = \mathbf{\$9.84 / \text{month}}\quad (\mathbf{₹942 / \text{month}}$ total company bill).

3. Billing Strategy: Freemium Activation Engine

- **Activation Hook: First 5 campaign setups 100% FREE** for all accounts. In India, charging non technical users for setup assistance creates instant hesitation, onboarding drop-offs, and negative reviews.
- **Monetization (Campaign 6 Onward):** Once an account demonstrates power usage, offer a **Pro Copilot Add-On (₹499/month)** or bundle it into Cheerio's **Growth Plan (₹2,499/month)**.
- **Margin:** At ~₹0.19/query, 15 campaigns cost Cheerio ₹2.85 against a ₹499 fee, delivering a **99.4% gross margin**.

4. Monthly Auditing & Profitability Controls

17. **Daily Token Spike Alert:** Automated ceiling alert at **$2.00/day** on Groq (~₹190/day) to catch runaway loops or DDoS attempts.
18. **15-Turn Session Throttling:** Cap conversations at 15 turns/hour; automatically trigger friendly human support handover before user frustration builds.
19. **Weekly Monday Escalation Audit:** Review the 40% escalated tickets weekly to identify recurring policy/integration blockers and patch Copilot prompts within 24 hours.

5. Human Support Cost vs. AI Savings

- **Human Support Baseline:** Tier-1 agent cost = **₹30,000/month**; capacity = **500 tickets/month** (₹60.00/ticket).
- 5,000 queries require **10 agents** = **₹3,00,000 / month** (₹3.0 Lakhs).
- **With AI Copilot:** Human team handles only the 2,000 escalated queries = **4 agents** (₹1,20,000/month) + ₹942 Groq AI cost = **₹1,20,942 / month**.
- **Net Financial Impact:**
- **Monthly Net Savings:** $₹3,00,000 - ₹1,20,942 = \mathbf{₹1,79,058 / \text{month}}$ (~**₹1.79 Lakhs/month**).
- **Annual Net Savings:** $₹1,79,058 \times 12 = \mathbf{₹21,48,696 / \text{year}}$ (~**₹21.5 Lakhs/year**).
- **Cost Reduction: 59.7% reduction** in overall support expenditure.

6. Complete Unit Economics Model (Formulas & Assumptions)

Metric / Parameter	Value / Formula	Unit	Notes / Benchmarks
Active Cheerio Businesses	5,000	Businesses	Given in assignment brief
Monthly Queries per Business	1.0	Queries/mo	Blended rate across active & passive accounts
Total Monthly Queries	5,000 (5,000 × 1.0)	Queries/mo	Total incoming setup & delivery tickets
AI Self-Serve Resolution Rate	60%	%	3-month North Star target
Queries Resolved by AI	3,000 (5,000 × 60%)	Queries/mo	Resolved self-serve without human agent
Queries Escalated to Human	2,000 (5,000 - 3,000)	Queries/mo	Complex edge cases & policy appeals

Input Token Price per 1k	$0.00059	USD ($)	Groq Enterprise Llama 3 70B ($0.59 / 1M)
Output Token Price per 1k	$0.00079	USD ($)	Groq Enterprise Llama 3 70B ($0.79 / 1M)
Input Tokens per Query	3,000	Tokens	Lean prompt: system + state + pruned history
Output Tokens per Query	250	Tokens	Concise suggestions & JSON form updates
AI Cost per Query (USD)	$0.0019675	USD ($)	Input cost ($0.00177) + Output cost ($0.0001975)
Total Monthly AI Cost (USD)	$9.84	USD ($)	5,000 queries × $0.0019675
USD to INR Exchange Rate	95.76	INR (₹)	Live mid-market exchange rate
Monthly AI Cost (INR)	**₹942 / month**	INR (₹)	Total company AI bill (office coffee budget)
Support Agent Monthly Cost	₹30,000	INR (₹)	Tier-1 support agent salary baseline
Tickets / Agent / Month	500	Tickets	~22–23 tickets/day across 22 working days
Cost per Human Ticket	₹60.00	INR (₹)	₹30,000 salary / 500 tickets
Agents Needed (Baseline)	10 agents	Agents	5,000 tickets / 500 (without AI)
Baseline Human Support Cost	₹3,00,000 / month	INR (₹)	10 agents × ₹30,000 (₹3.0 Lakhs)
Agents Needed (With AI)	4 agents	Agents	2,000 escalated tickets / 500 (saves 6 FTEs)
New Total Support Cost	₹1,20,942 / month	INR (₹)	4 agents (₹1.20L) + Groq AI cost (₹942)
Net Monthly Savings	**₹1,79,058 / month**	**INR (₹)**	**Baseline - New Cost (~₹1.79 Lakhs/mo)**
Net Annual Savings	**₹21,48,696 / year**	**INR (₹)**	**Monthly Savings × 12 (~₹21.5 Lakhs/yr)**
Support Cost Reduction	**59.7% reduction**	**%**	**Direct reduction in support expenditure**